MT8 – Yes/No Audio Classification

Digital Speech Processing | Mini-Task Report

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Implementation and reported measurements are based on the executed Colab notebook supplied for this assignment.

1. Introduction

In this mini-task, I used TorchAudio's YESNO speech dataset to compare four binary audio classifiers. I first inspected the audio signals and their labels, then separated candidate spoken words, extracted acoustic features, and trained a Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Random Forest, and a Multi-Layer Perceptron (MLP) neural network. I evaluated the models using the same held-out recordings and a group-based cross-validation check.

2. Dataset and Initial Inspection

The dataset loader returned 60 mono recordings with eight binary labels per recording (0 = No and 1 = Yes). The first recording contained 50,800 samples at 8,000 Hz, corresponding to 6.35 seconds. Its labels were [0, 0, 0, 0, 1, 1, 1, 1]. Each original file contains a sequence of responses, so I did not treat a complete recording as a single Yes/No sample.

Property	Observed value
Original recordings	60
Sampling rate	8,000 Hz
Channels	1 (mono)
Labels per recording	8
First-recording duration	6.35 s

3. Waveform and Spectrogram

I plotted the waveform and spectrogram of the first recording to inspect the timing and spectral content of its speech regions. The displays showed several separated regions of activity, which provided a starting point for word segmentation; the figures alone do not establish exact word boundaries.

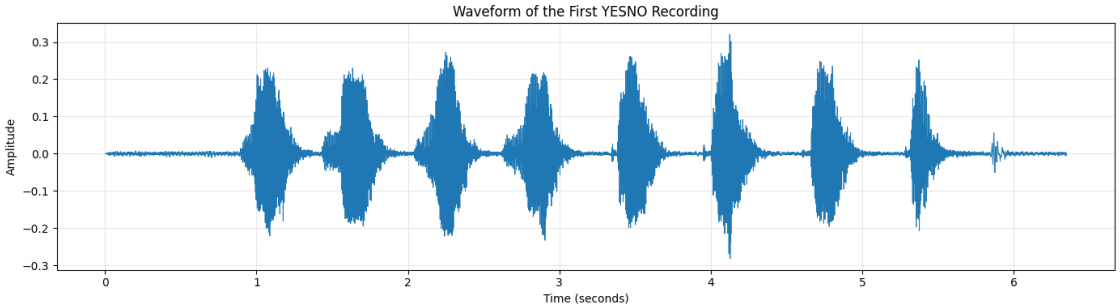


Figure 1. Waveform of the first YESNO recording (executed notebook).

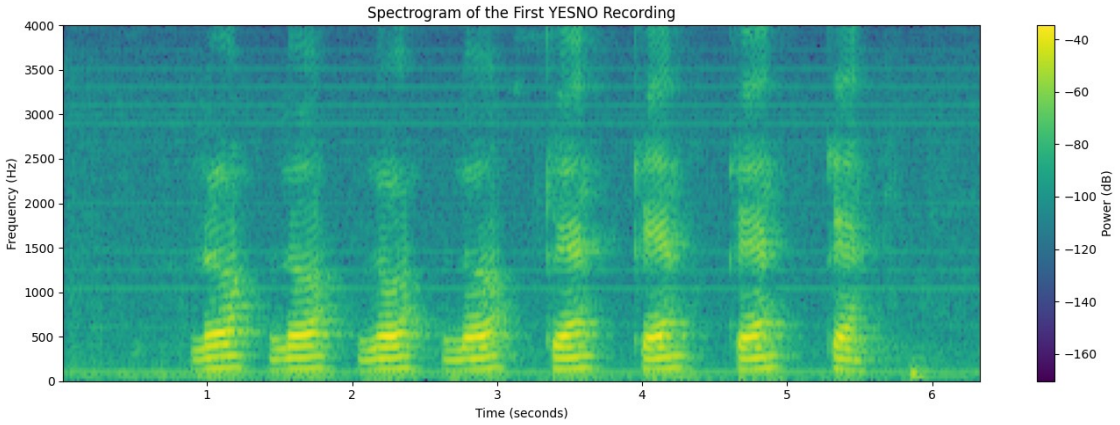


Figure 2. Spectrogram of the same recording (executed notebook).

4. Segmentation and Data Preparation

I used a smoothed signal-energy threshold and morphological operations to identify speech regions. The first parameter configuration returned the expected eight regions in 55 of 60 recordings. After lowering the threshold and relaxing the duration and silence settings, 59 of 60 recordings matched the expected count. Recording 21 still produced seven regions against eight labels, so I excluded it rather than assigning uncertain labels. Count agreement alone does not prove that every word boundary and label is correct.

From the remaining 59 recordings, I extracted 472 candidate word segments with up to 50 ms padding on either side. The class counts were 223 No and 249 Yes, and the segment durations ranged from 0.204250 to 0.689375 seconds (mean 0.345753 seconds). There were no missing values in the generated metadata.

5. Feature Extraction and Experimental Setup

For each segment, I computed 13 MFCCs, zero-crossing rate, RMS energy, and spectral centroid. Means and standard deviations yielded a 32-dimensional feature vector (26 MFCC statistics and six statistics from the other measures). The final feature matrix contained 472 rows and 32 columns, with no NaN or infinite values.

I split the data by source recording using GroupShuffleSplit (random_state = 42). The training portion contained 376 segments from 47 recordings, while the test portion contained 96 segments from 12 different recordings; no recording ID appeared in both sets. StandardScaler was fitted within the SVM, KNN, and MLP training pipelines, and was not fitted on the held-out test set.

Partition	Recordings	No	Yes	Total segments
Training	47	180	196	376
Testing	12	43	53	96
Total	59	223	249	472

6. Classification Models

I used an RBF-kernel SVM (C = 1.0, gamma = scale), KNN (k = 5), Random Forest (100 trees, random_state = 42), and an MLP with two hidden layers of 64 and 32 units (ReLU activation, maximum 1,000 iterations, random_state = 42). I applied the same training and test split to all four methods. These were fixed experimental configurations rather than parameters optimized against the test set.

7. Results

All four classifiers correctly predicted the 96 test segments according to the available automatic segment-label mapping. Table 3 reports the metrics and measured training times from the re-executed notebook.

Model	Accuracy	Precision	Recall	F1	Fit time (s)
SVM	1.0000	1.0000	1.0000	1.0000	0.017912
KNN	1.0000	1.0000	1.0000	1.0000	0.005055
Random Forest	1.0000	1.0000	1.0000	1.0000	0.413030
MLP Neural Network	1.0000	1.0000	1.0000	1.0000	0.163639

The four confusion matrices were identical: 43 correctly predicted No segments and 53 correctly predicted Yes segments, with no errors in the test partition. Figure 3 shows the SVM matrix as a representative example; the other three are available in the notebook.

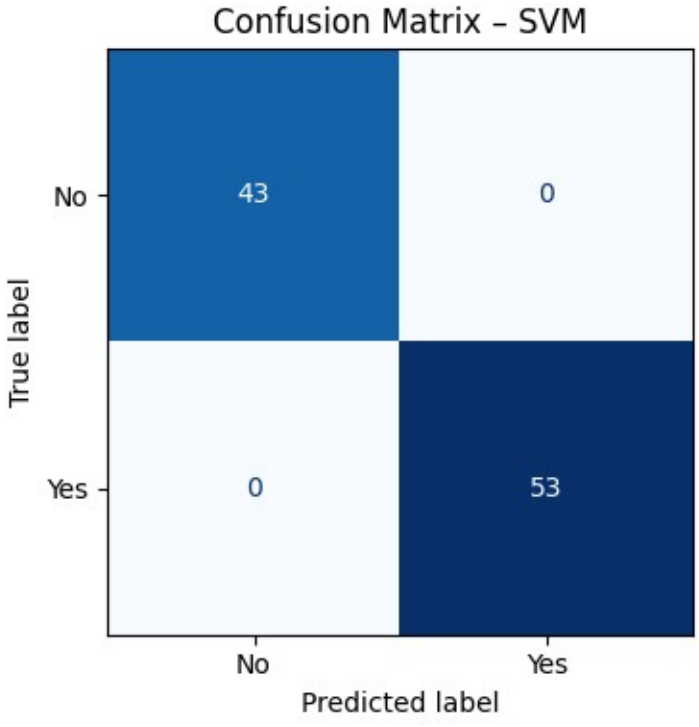


Figure 3. SVM confusion matrix; the other three classifiers produced the same counts.

Five-fold GroupKFold cross-validation on the training partition returned accuracy = 1.0000 and F1 = 1.0000 in every fold for all four classifiers (mean 1.0000, standard deviation 0.0000). This check kept samples from the same recording in the same fold and did not reuse the held-out test recordings for fitting.

8. Discussion

In this experiment, the four classifiers were tied on test accuracy and F1-score, so the results do not identify one as more accurate. KNN recorded the shortest fit time, but training time alone does not compare total computational cost, because KNN can require more work at prediction time. The timing values are specific to this CPU execution environment.

The perfect scores require careful interpretation. The labels attached to the automatically segmented word regions were not independently verified for every segment; matching eight detected regions to eight supplied labels is only a count-based check. I excluded recording 21 because its segmentation was ambiguous. The dataset is also small and contains one speaker, so this experiment does not establish generalization to unseen speakers, new microphones, or noisier environments. The reported scores are therefore measurements for this particular dataset and evaluation protocol, not general performance guarantees.

9. Conclusion

I completed an end-to-end classification experiment using TorchAudio's YESNO dataset. I prepared candidate word segments, built 32-dimensional acoustic feature vectors, and compared SVM, KNN, Random Forest, and MLP classifiers on a recording-disjoint test split. All four obtained 1.0000 accuracy and F1-score on the 96 tested segments, with identical confusion matrices. An important next improvement would be to verify word boundaries and label alignment independently and evaluate the methods on additional speakers.

References

[1] PyTorch Audio documentation, https://docs.pytorch.org/audio/stable/index.html

[2] OpenSLR, YESNO corpus (SLR1), https://www.openslr.org/1/

[3] Executed Google Colab notebook (source of the reported experimental figures and numeric results): https://colab.research.google.com/drive/1VlwltIJVzuLK5MPepdL3-r5yJC88wVIh?usp=sharing